

## Counterfactual Analysis: Difference-in-Differences (DiD) Estimation

To estimate the true financial impact of the mid-year targeting strategy change, a Difference-in-Differences (DiD) methodology was applied. This approach isolates the causal effect of the new strategy by subtracting the natural, underlying baseline shifts observed in the legacy accounts.

### Methodology & Cohort Definitions

- **Identification Strategy:** Difference-in-Differences (DiD) applied exclusively to operational revenue (payments made within the strict 72-hour attribution window).
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- **Treatment Group:** Accounts actively subjected to the new targeting strategy on or after the June 1, 2026 cutoff date.
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- **Control Group:** A randomized holdout group of accounts that remained on the legacy targeting strategy during the identical timeframe.
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### The DiD Results

- The Control group saw a natural baseline increase of ₹3,601.54 per account during the post-intervention period.
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- The Treatment group saw an increase of ₹8,083.90 per account during the same period.
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- **DiD Estimator:** The net operational impact of the new targeting strategy is **+₹4,482.36** per account (₹8,083.90 - ₹3,601.54).
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### Statistical Constraints & Correct Reasoning

While the mathematical estimator suggests a positive return, the validity of this counterfactual relies entirely on the underlying assumptions, which expose critical flaws in the data.

- **Assumptions:** DiD strictly relies on the **Parallel Trends Assumption**, which dictates that in the absence of the strategy change, the Treatment group's revenue would have trended exactly parallel to the Control group.
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- **Confounding Factors:** Natural seasonality, organic self-curing payments (partially mitigated by the 72-hour filter), and late-arriving data bias in recent cohorts (mitigated by dropping August data).
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- **Limitations (Assumption Violation):** The visual "Parallel Trends Check" definitively proves that the two groups were *not* moving in parallel prior to the June 1st intervention.

Specifically, the Treatment group exhibited a massive, unexplainable collapse in April compared to the Control group. Because the pre-intervention trends are highly divergent, the basic DiD estimator of +₹4,482.36 is mathematically biased. To extract a truly valid counterfactual, Propensity Score Matching (PSM) must be applied to synthetically balance the accounts based on their starting characteristics before recalculating the DiD. Furthermore, because previous forensics proved the core demographic metrics (like **risk\_segment**) are synthetic, any targeting strategy utilizing those metrics is fundamentally flawed from inception.